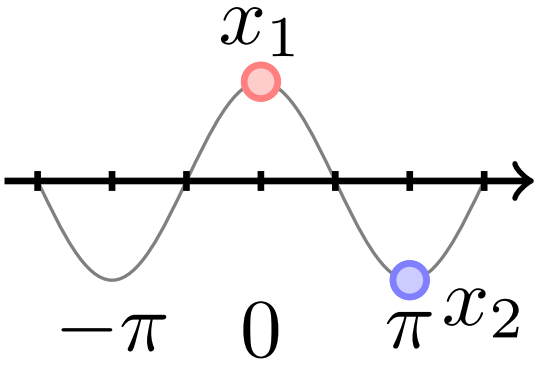
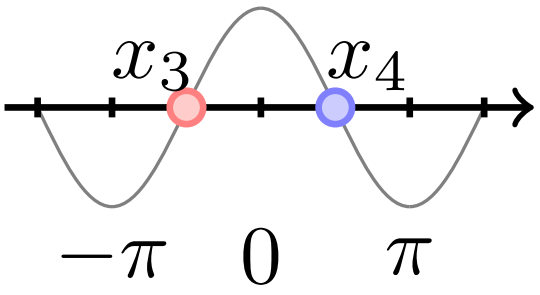


$$x_1, x_2 \sim \mathcal{X}_1 : ||x_1 - x_2||_1 = \pi$$

$$x \sim \mathcal{X}_1 : \mathbb{E}[x] = \frac{\pi}{2} \wedge \mathrm{Var}[x] = \frac{\pi}{2}$$



$$x_3, x_4 \sim \mathcal{X}_2 : ||x_3 - x_4||_1 = \pi$$

$$x \sim \mathcal{X}_2 : \mathbb{E}[x] = 0 \wedge \mathrm{Var}[x] = \frac{\pi}{2}$$